ATTENTION-FREE KEYWORD SPOTTING

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ABSTRACT

Till now, attention-based models have been used with great success in the keyword spotting problem domain. However, in light of recent advances in deep learning, the question arises whether self-attention is truly irreplaceable for recognizing speech keywords. We thus explore the usage of gated MLPs—previously shown to be alternatives to transformers in vision tasks—for the keyword spotting task. We provide a family of highly efficient MLP-based models for keyword spotting, with less than 0.5 million parameters. We show that our approach achieves competitive performance on Google Speech Commands V2-12 and V2-35 benchmarks with much fewer parameters than self-attention-based methods.

1 INTRODUCTION

Transformers (Vaswani et al., 2017) have been a highly disruptive innovation in deep learning. They were originally introduced in the domain of Natural Language Processing, and have since replaced classical recurrent neural networks as the default approach in many NLP tasks (Devlin et al., 2018; Yang et al., 2019). Not only have Transformers transformed the field of NLP, but they have also been proven to have competitive performance across several important problem domains—such as image classification (Dosovitskiy et al., 2020; Yuan et al., 2021), video classification (Neimark et al., 2021), object detection (Carion et al., 2020), automatic speech recognition (Liu et al., 2021a; Gulati et al., 2020) and so on.

Partly due to their remarkable success in vision tasks, Transformers have lately been studied in the field of keyword spotting (KWS) (Berg et al., 2021; Gong et al., 2021), and have shown similar exceptional results. Recent research (Tolstikhin et al., 2021; Liu et al., 2021b; Melas-Kyriazi, 2021) shows that a core component of Transformers, self-attention, may not be necessary for achieving good performance in vision and language tasks. This finding necessitates a study on whether MLPs can be an alternative to self-attention, which has been a main focus of several recent state-of-the-art methods for the KWS problem. Our contributions can be summarized as follows:

1. We introduce the Keyword-MLP (KW-MLP), an efficient, attention-free alternative to the Keyword Transformer (KWT) (Berg et al., 2021). It achieves 97.63% and 97.56% accuracy on the Google Speech Commands V2-12 and V2-35 benchmarks (Warden, 2018) respectively—showing comparable performance to the KWT, while having much fewer parameters.
2. We distill smaller and shallower versions of KW-MLP, with the smallest having 0.213 million parameters, and accuracies of 97.12% and 97.17% on Google Speech Commands V2-12 and V2-35 benchmarks respectively.

2 RELATED WORK

2.1 KEYWORD SPOTTING

Keyword spotting (KWS) deals with identifying some pre-specified speech keywords from an audio stream. As it is commonly used in always-on edge applications, KWS research often focuses on

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both accuracy and efficiency. While research in keyword spotting goes back to the 1960s (Teacher et al., 1967), most of the recent and relevant works have been focused on the Google Speech Commands dataset (Warden, 2018), which has inspired numerous works and has rapidly grown to be the standard benchmark in this field. The dataset contains 1 second long audio clips, each containing an utterance of a word. Notably, there are two versions, V1 and V2, consisting of 30 and 35 keywords respectively. There is also a 12 keyword task for either version, where it is required to identify 10 keywords and two additional classes, 'silence' and 'unknown' (containing instances of the unused keywords.)

Initial approaches to keyword spotting on the speech commands consisted of convolutional models (Warden, 2018). Majumdar & Ginsburg (2020), Mordido et al. (2021) and Zhang & Koishida (2017) proposed lightweight CNN models with depth-wise separable convolutions. de Andrade et al. (2018) proposed using a convolutional recurrent model with attention, introducing the usage of attention in the KWS task. Rybakov et al. (2020) proposed a multi-headed, self-attention-based RNN (MHAtt-RNN). Vygon & Mikhaylovskiy (2021) proposed an efficient representation learning method with triplet loss for KWS. While the state of the art in KWS at that time was the method of Rybakov et al. (2020), it was empirically seen that triplet loss performed poorly with RNN-based models. The authors later obtained excellent results with ResNet (Tang & Lin, 2018) variants.

Recently, Berg et al. (2021) and Gong et al. (2021) proposed the Keyword Transformer (KWT) and Audio Spectrogram Transformer (AST) respectively. Both approaches are inspired by the success of the Vision Transformer (ViT) (Dosovitskiy et al., 2020), and show that patch-based transformer models with self-attention can obtain state of the art or comparable results on the keyword spotting task. A key difference between these two transformer approaches is that AST uses ImageNet (Deng et al., 2009) and Audioset (Gemmeke et al., 2017) pre-training. Furthermore, the best performing KWT models are trained with attention-based distillation method (Touvron et al., 2021) using a teacher MHAtt-RNN (Rybakov et al., 2020).

2.2 MLP-based Vision

The Vision Transformer (Dosovitskiy et al., 2020) has thus far shown the remarkable capability of Transformers on image and vision tasks. However, several recent works have questioned the necessity of self-attention in ViT. Melas-Kyriazi (2021) directly raises the question on the necessity of attention, and shows that the effectiveness of the Vision Transformer may be more related to the idea of the patch embedding rather than self-attention. Tolstikhin et al. (2021) proposed the MLP-Mixer, which performs token mixing and channel mixing on image patches/tokens, and shows competitive performance on the ImageNet benchmark. Liu et al. (2021b) proposed the gMLP, consisting of very simple channel projections and spatial projections with multiplicative gating—showing remarkable performance without any apparent use of self-attention.

3 Keyword-MLP

Inputs to KW-MLP consist of mel-frequency cepstrum coefficients (MFCC). Let an arbitrary input MFCC be denoted as $\boldsymbol{X} \in \mathbb{R}^{F \times T}$, where F and T are the frequency bins and time-steps respectively. We divide $\boldsymbol{X}$ into patches of shape $F \times 1$, getting a total of T patches. Each patch is effectively a vector of mel-frequencies for a particular time-step.

The T patches are flattened, giving us $\boldsymbol{X}_0 \in \mathbb{R}^{T \times F}$. We then map $\boldsymbol{X}_0$ to a higher dimension d, with a linear projection matrix $\boldsymbol{P}_0 \in \mathbb{R}^{F \times d}$, getting the frequency domain patch embeddings $\boldsymbol{X}_E$.

$$\boldsymbol{X}_E = \boldsymbol{X}_0 \boldsymbol{P}_0 \tag{1}$$

The obtained $\boldsymbol{X}_E$ is passed through L consecutive, identical gated-MLP (gMLP) blocks (Liu et al., 2021b). On a high level, we can summarize the gMLP blocks used in KW-MLP as a pair of projections across the embedding dimension separated by a projection across the temporal dimension. The block can also be formulated with the following set of equations (omitting bias and normalization for the sake of conciseness):

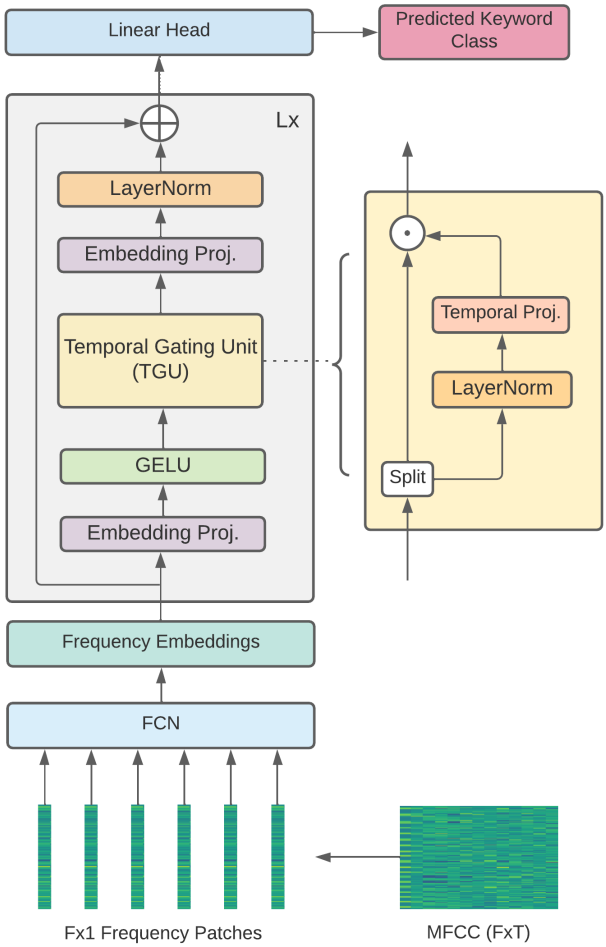

Figure 1: The Keyword-MLP architecture. It consists of L consecutive gMLP blocks. Note that $\oplus$ represents the typical added skip-connection, while $\odot$ represents element-wise product.

$$\begin{aligned} \boldsymbol{Z} &= \sigma(\boldsymbol{X}_{in}\boldsymbol{U}) \\ \tilde{\boldsymbol{Z}} = g(\boldsymbol{Z}) &= g([\boldsymbol{Z}_r \boldsymbol{Z}_g]) = \boldsymbol{Z}_r \odot (\boldsymbol{Z}_g^T \boldsymbol{G})^T \\ \boldsymbol{X}_{out} &= \tilde{\boldsymbol{Z}}\boldsymbol{V} \oplus \boldsymbol{X}_{in} \end{aligned} \tag{2}$$

First, we use the matrix $\boldsymbol{U} \in \mathbb{R}^{d \times D}$ to linearly project $\boldsymbol{X}_{in}$ from the embedding dimension d to the projection dimension D (essentially a matmul operation). σ represents the GELU activation function. $g(Z)$ represents the the Temporal Gating Unit (TGU) shown in Figure 1. The input to TGU, $\boldsymbol{Z}$, is first split into $\boldsymbol{Z}_r$ and $\boldsymbol{Z}_g \in \mathbb{R}^{T \times D/2}$, the residual and the gate respectively. We use the matrix $\boldsymbol{G} \in \mathbb{R}^{T \times T}$ to performs the linear projection across the temporal axis. This is followed by the linear gating—an element-wise multiplication with the residual $\boldsymbol{Z}_r$. While the temporal projection operation can be implemented as passing $\boldsymbol{Z}_g$ transposed through a `Dense(T, T)` layer, in practice, it can be implemented more efficiently by instead passing $\boldsymbol{Z}_g$ through a `Conv1D(T, T, 1)` layer. $\tilde{\boldsymbol{Z}} \in \mathbb{R}^{T \times D/2}$ is projected back to the embedding dimension d with the matrix $\boldsymbol{V}$ and then added with the skip-connected input $\boldsymbol{X}_{in}$.

The original gMLP paper (Liu et al., 2021b) applies LayerNorm *before* the initial channel projection (analogous to embedding projections for images). We however find that applying norm *after* the second embedding projection results in a notably faster and more optimal convergence. Berg et al. (2021) also observe a similar phenomenon in their work.

The overall system is shown in Figure 1. In KW-MLP, we primarily use $L = 12$ (12 consecutive gMLP blocks), embedding dim $d = 64$, and projection dim $D = 256$. We also explore a group of smaller KW-MLP models with shallower depth, i.e. $L = 10, 8, 6$. The input MFCCs to the model are of shape 40×98, where 40 is the number of frequency bins, and 98 is the number of timesteps. All settings are also shown in Table 2.

It is to be noted that the largest KW-MLP model has only 0.424M parameters, which is smaller than the smallest KWT variant (KWT-1 with 0.607M params) and much smaller than AST models (87M params). However, from Table 1 we can see that KW-MLP shows competitive accuracy with these models, particularly on the Speech Commands V2-35 benchmark.

3.1 Knowledge Distillation

In order to boost the accuracies of shallower KW-MLP variants, we use knowledge distillation (KD) (Hinton et al., 2015), using the KW-MLP model with $L = 12$ as the teacher model. We use an annealed KD approach (Jafari et al., 2021) where the temperature parameter decreases every step following a cosine-annealing rule till it reaches 1. Other KD parameters, such as alpha, are shown in Table 2. We also do not use label smoothing when training with KD, as soft targets are obtained from teacher predictions.

Table 1: Comparison of Model Parameters and Accuracy on Google Speech Commands V2-12 and V2-35 benchmarks (Warden, 2018)

Method	Extra Knowledge	V2-12	V2-35	# Params (M)
Att-RNN [3]		96.9	93.9	0.202
Res-15 [25]		98.0	96.4	0.237
MHAtt-RNN [19]		98.0	97.27	0.743
AST-S [8]	Pre. ImageNet		**98.11**	87
AST-P [8]	Pre. ImageNet & Audioset		97.88	87
KWT-3 [1]	KD with MHAtt-RNN	**98.56**	97.69	5.361
KWT-2 [1]	KD with MHAtt-RNN	98.43	97.74	2.394
KWT-1 [1]	KD with MHAtt-RNN	98.08	96.95	0.607
KWT-3 [1]		98.54	97.51	5.361
KWT-2 [1]		98.21	97.53	2.394
KWT-1 [1]		97.72	96.85	0.607
KW-MLP		**97.63**	**97.56**	0.424
KW-MLP$_{10}$		97.38	97.35	0.353
KW-MLP$_{8}$		97.28	97.26	0.283
KW-MLP$_{6}$		97.03	97.07	0.213
KW-MLP$_{10}$	KD with KW-MLP	97.30	97.49	0.353
KW-MLP$_{8}$	KD with KW-MLP	97.24	97.45	0.283
KW-MLP$_{6}$	KD with KW-MLP	97.12	97.17	0.213

4 Experimental Details

We follow similar hyperparameters to Rybakov et al. (2020); Berg et al. (2021), with minor changes; all our hyperparameters and settings are shown in Table 2. For training, we use a smaller batch-size of 256, and train for 140 epochs. No other augmentation apart from Spectral Augmentation (Park et al., 2019) is applied (to enable fast training). As an additional regularization method, each gMLP block has a survival probability of 0.9 (alternatively, a 0.1 probability to drop each block). We run experiments on Google Speech Commands V2-12 and V2-35 benchmarks, following the standard protocol described in Warden (2018).

As seen from Table 1, the largest KW-MLP model has only 424K parameters, which is much fewer than the KWT models, while having comparable accuracy. Furthermore, since we do not apply expensive run-time augmentations like resampling, time-shifting, adding background noise, mixup, etc. (used by (Rybakov et al., 2020; Berg et al., 2021; Gong et al., 2021)), it is possible to train KW-MLP models in a very short time on free cloud compute such as the NVIDIA Tesla K80 or Tesla P100 provided by Google Colab and Kaggle.

As a trade-off for fast training, a limitation of the KW-MLP experiments is that the effect of various augmentation methods have not been explored. This is more apparent in the V2-12 task, which contains much fewer training examples (~ 37000) than the V2-35 task (~ 84000). The KW-MLP model does not generalize as well here, as compared to V2-35.

Table 2: Overview of Hyper-Parameters and Settings

Training		Augmentation		Model	
Epochs	140	# Time Masks	2	# Blocks, L	12
Batch Size	256	Time Mask Width	[0, 25]	Input Shape	40×98
Optimizer	AdamW	# Freq Masks	2	Patch Size	40×1
Learning Rate	0.001	Freq Mask Width	[0, 7]	Dim, d	64
Warmup Epochs	10			Dim Proj.	256
Scheduling	Cosine			# Classes	35
Regularization		**Audio Processing**		**KD**	
Label Smoothing	0.1	Sampling Rate	16000	α	0.9
Weight Decay	0.1	Window Length	30 ms	Init Temp	5.0
Block Survival Prob.	0.9	Hop Length	10 ms		
		n_mfcc	40		

5 Temporal Projection Matrices

In Figure 2, we visualize the weights of the temporal gating unit, i.e. the temporal projection matrix $\boldsymbol{G} \in \mathbb{R}^{T \times T}$ in equation 2. Interestingly, we can observe that our model learns weights similar to Toeplitz matrices. This suggests that KW-MLP learns a form of shift-invariance, which is necessary for the keyword spotting task. For instance, in a 1 second audio clip, a keyword can occur at different temporal positions; so the model needs to be invariant to temporal shift.

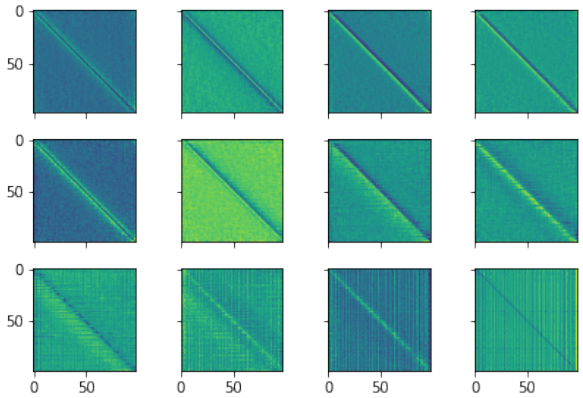

Figure 2: Visualization of the temporal projection matrices, $\boldsymbol{G}$, for each of the $L = 12$ gMLP blocks of KW-MLP. The matrices are arranged in row major order.

6 Conclusion

The Keyword-MLP has shown itself to be an efficient solution to the keyword spotting task, and an alternative to self-attention-based methods. We hope that we provide an additional avenue of future research in audio and speech domains, particularly when resource-efficiency is concerned.

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